

Numpy 4 Hacker Rank

1.

Task

You are given two arrays A and B . Both have dimensions of $N \times N$.

Your task is to compute their [matrix product](#).

Input Format

The first line contains the integer N .

The next N lines contains N space separated integers of array A .

The following N lines contains N space separated integers of array B .

Output Format

Print the matrix multiplication of A and B .

Sample Input

```
2
1 2
3 4
1 2
3 4
```

Sample Output

```
[[ 7 10]
 [15 22]]
```

```
1  # Enter your code here. Read input from STDIN. Print output to STDOUT
2  import numpy as np
3
4  n = int(input())
5
6  arr=[]
7  arr1=[]
8
9  for _ in range(n):
10     arr.append(list(map(int,input().strip().split())))
11
12  for _ in range(n):
13     arr1.append(list(map(int,input().strip().split())))
14
15
16  a = np.array(arr)
17  b = np.array(arr1)
18
19  print(np.dot(a,b))
```

Exit Full Screen View



You have earned 20.00 points!

You are now 5 points away from the gold level for your python badge.

97%

395/400

Congratulations

You solved this challenge. Would you like to challenge your friends?   

[Next Challenge](#)

✓ Test case 0

✓ Test case 1 

✓ Test case 2 

Compiler Message

Success

Input (stdin)

1	2
2	1 2
3	3 4
4	1 2
5	3 4

Expected Output

1	[[7 10]
2	[15 22]]

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2.

Task

You are given two arrays: A and B .

Your task is to compute their inner and outer product.

Input Format

The first line contains the space separated elements of array A .

The second line contains the space separated elements of array B .

Output Format

First, print the inner product.

Second, print the outer product.

Sample Input

```
0 1
2 3
```

Sample Output

```
3
[[0 0]
 [2 3]]
```

```
1 # Enter your code here. Read input from STDIN. Print output to STDOUT
2 import numpy as np
3 a = np.array(list(map(int,input().split())))
4 b = np.array(list(map(int,input().split())))
5 print(np.inner(a,b))
6 print(np.outer(a,b))
```



You have earned 20.00 points!
28/115 challenges solved.



Congratulations

You solved this challenge. Would you like to challenge your friends?

[Next Challenge](#)

Test case 0

Compiler Message

Test case 1

Success

Test case 2

Input (stdin)

[Download](#)

```
1 0 1
2 2 3
```

Expected Output

[Download](#)

```
1 3
2 [[0 0]
3 [2 3]]
```